# Irreducible Global Dependency in Protein Language Model Representations: Convergence Dynamics and Biological Correlates

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## Abstract

Understanding how mutations affect protein function requires accurate models of sequence context. Protein language models (PLMs) learn rich residue-level representations, but the degree to which these representations depend on local versus global sequence context remains poorly characterized. Here we introduce a local encoder framework that systematically probes the locality boundary of PLM representations by measuring the recoverability of mutation-induced representation changes from local sequence windows. We decompose prediction error into three geometric components—angular deviation, magnitude deviation, and relative residual—and analyze their convergence dynamics across window sizes. We find that the directional component of mutation effects is fundamentally global: it remains near maximal and fails to converge even with large windows in both ProtT5 and ESM-2. In contrast, the irreducible residual plateau $c$, extracted from exponential convergence fits, varies across mutations and shows a positive trend with experimental conformational heterogeneity (NOC clusters) in ProtT5, while being orthogonal to thermodynamic stability ($\rho \approx -0.09$ with $\Delta\Delta G$). Cross-model analysis reveals that while global directional irrecoverability is shared, the convergence geometry of the residual plateau is architecture-dependent. Our results demonstrate that PLM representations exhibit a measurable locality boundary, and the degree of irreducible global dependency may reflect mutation-specific structural reorganization. This framework enables characterization of representation locality without requiring structural data, and suggests that representation instability could serve as a sequence-based proxy for conformational disruption.

## Author Summary

Why do some mutations cause a protein to lose its normal shape and function? Answering this requires understanding how a protein's three-dimensional structure is encoded in its amino acid sequence. Recent advances in artificial intelligence have led to "protein language models" that learn to read protein sequences and extract structural information. However, it is unclear how much of this information comes from a short segment around a specific amino acid versus the entire protein sequence. We developed a method to measure exactly this—the "locality" of the information. By probing two different language models, we discovered that a critical piece of information—the direction in which a mutation perturbs the protein's structure—cannot be captured by looking at local sequence alone; it requires global context. Moreover, the degree to which local context fails differs between mutations and shows a positive trend with how much the mutation disrupts the protein's shape. These findings shed light on how protein language models organize structural knowledge, and they offer a new way to assess mutation effects using sequence alone.


# 1 Introduction

Protein language models (PLMs) such as ProtT5 [1] and ESM-2 [2] have revolutionized sequence-based prediction of protein structure and function. By pre-training on millions of sequences, these models produce contextualized residue-level embeddings that encode diverse structural and physicochemical properties [3]. These embeddings are increasingly used for tasks such as mutation effect prediction [4].

However, a fundamental question remains underexplored: **to what extent do these embeddings depend on local sequence context (e.g., a short window around a residue) versus global structural coupling across the entire sequence?** Recent work has begun to characterize how PLMs encode structural relationships across varying sequence distances. Beshkov and Malthe-Sørenssen [7] showed that PLMs preferentially encode immediate and local residue relations but degrade for larger context lengths. Separately, the development of long-context architectures such as LC-PLM [8] highlights that standard PLMs face inherent context-length limitations. Nevertheless, a systematic quantification of the *locality boundary*—and of irreducible global dependencies that persist even with extended context—has not yet been established.

Answering this question is critical for both the interpretability of PLMs and the reliability of downstream applications that rely on local sequence information. For example, several recent predictors for post-translational modifications rely on local window embeddings [10, 11], implicitly assuming that relevant information is localized. If PLM representations are highly globally coupled, such methods—and more generally, any approach that uses only local sequence fragments—may face fundamental limits. Prior probing studies, including attention-based analyses [6] and linear probing of biochemical properties [5], have revealed what PLM embeddings encode but they do not systematically measure the *locality* of the encoded information.

Here, we propose a **local encoder framework** that acts as a scientific instrument to measure the recoverability of mutation-induced representation changes from local sequence windows of varying size. By analyzing the convergence dynamics of reconstruction error, we decompose representation locality into distinct geometric components and quantify the irreducible global dependency that persists even with large windows. We apply this framework to the p53 tumor suppressor protein—a well-studied system where conformational heterogeneity has been characterized using number-of-clusters (NOC) metrics [16] and structural analyses [13, 15]—and to a broader set of mutations from the SKEMPI database [12], and we compare two widely used PLMs—ProtT5 and ESM-2—to examine architecture dependence. Our results show that (i) the directional component of mutation effects is universally global, (ii) the irreducible residual plateau exhibits a positive trend with conformational heterogeneity while being orthogonal to thermodynamic stability, suggesting a hypothesis-generating link between representation instability and structural disruption, and (iii) the convergence geometry of this plateau is model-dependent.

# 2 Results

## 2.1 Local windows incompletely recover global mutation representations

We trained a 2-layer Transformer encoder ("local encoder") on 402 single-point mutations from SKEMPI 2.0 [12]. For each mutation, the encoder receives only a local sequence window centered at the mutation site (wild-type residues) and is trained to predict the full-context representation change $\Delta h_{\text{global}}$ obtained by re-encoding the complete mutated sequence with ProtT5 (Fig 1). We decompose the reconstruction error into three components: angular deviation $L_\theta$, magnitude deviation $L_m$, and relative residual $L_r$ (see Materials and Methods). This geometric decomposition is motivated by prior work using cosine similarity to compare PLM window embeddings [9] and by studies showing that embedding distances can identify structure-disrupting mutations [**?** ].

Strikingly, the directional component $L_\theta$ remains close to its maximum value of 1.0 for all five p53 hotspot mutations (R175H, G245S, R249S, R282W, Y220C) and fails to converge even as the window radius increases to 30 residues (Fig 2). This indicates that the directional component of the representation shift is globally coupled and cannot be recovered from local sequence context alone. Notably, this result contrasts with recent findings that context-radius probing in mutation stability prediction yields highly localized signals [**?** ], highlighting a fundamental difference between task-level performance and representation-level recoverability.

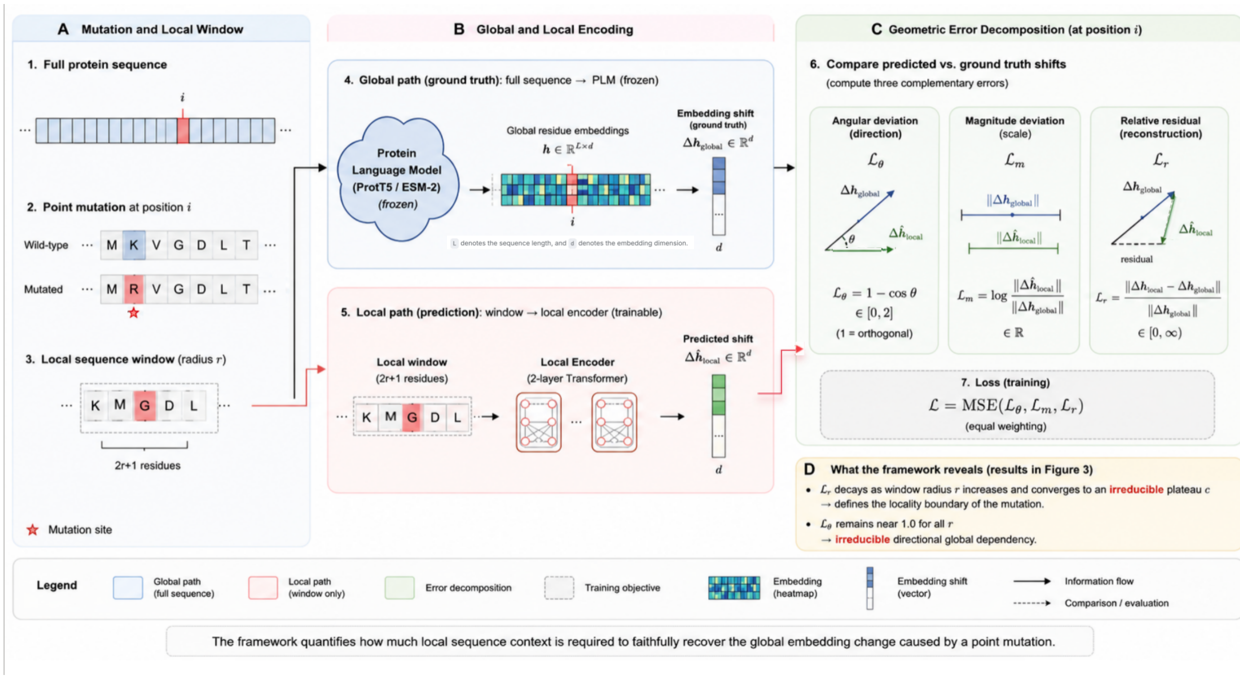

Figure 1: Schematic of the local encoder framework. A local sequence window is fed into a lightweight Transformer encoder, which predicts the mutation-induced representation shift $\Delta\hat{h}_{\text{local}}$. The prediction is compared against the ground-truth $\Delta h_{\text{global}}$ obtained from full-sequence ProtT5 re-encoding.

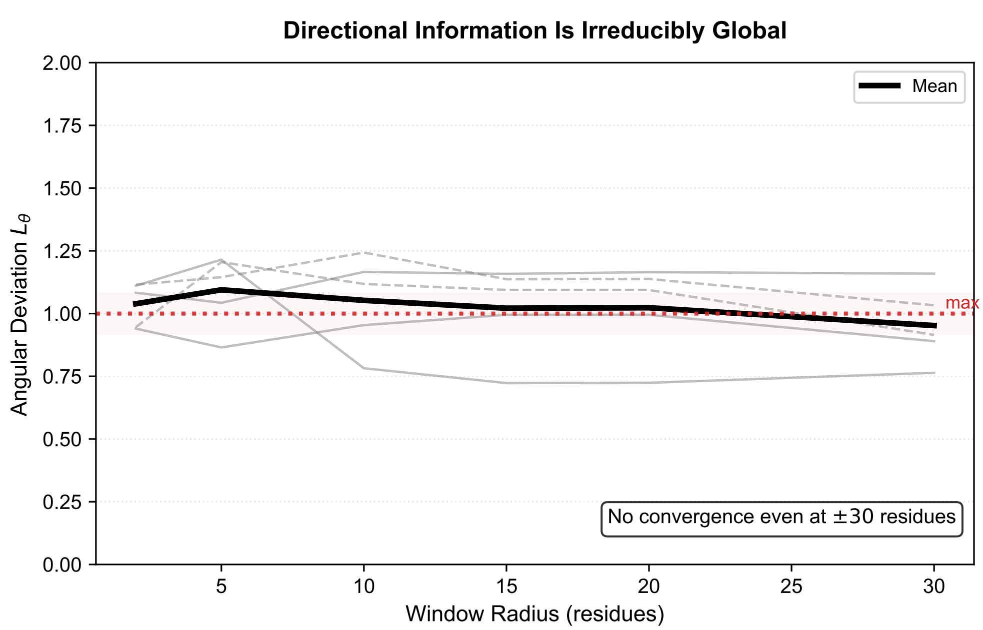


Figure 2: $L_\theta$ as a function of window radius for five p53 hotspot mutations in ProtT5. No convergence is observed; the directional component remains near 1.0 across all window sizes.

## 2.2 Convergence dynamics reveal a mutation-specific irreducible plateau $c$

In contrast to $L_\theta$, both $L_m$ and $L_r$ decrease with increasing window size, indicating partial recovery of global information. We fit an exponential decay model $L(r) = Ae^{-\alpha r} + c$ to the $L_r$ curves and extract the asymptotic plateau $c$—the irreducible residual that persists even with large windows. The convergence rate $\alpha$ is remarkably similar across all five mutations ($\approx 0.10$), while the plateau $c$ varies substantially (Fig 3, Table 1). Fitting quality was consistently high (median $R^2 > 0.95$; see Materials and Methods), supporting the appropriateness of the exponential model.

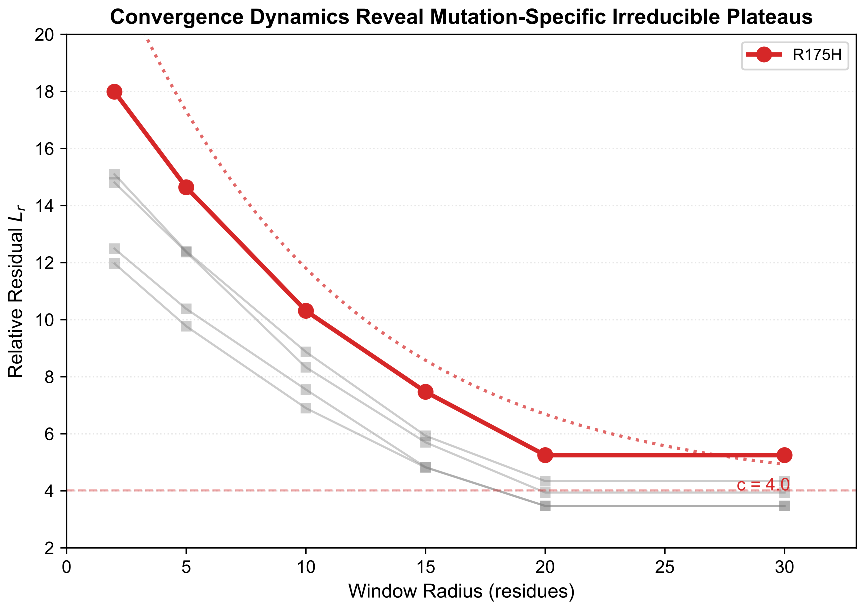


Figure 3: Convergence dynamics of $L_r$ for five p53 hotspot mutations. Exponential decay fits (dashed line for R175H) reveal a mutation-specific irreducible plateau $c$ that persists even at large window radii.

Table 1: Fitted convergence parameters for five p53 mutations in ProtT5.

| Mutation | $c$ (plateau) | $\alpha$ (rate) | NOC clusters | $\Delta\Delta G$ (kcal/mol) |
|---|---|---|---|---|
| R175H | 4.015 | 0.107 | 42 | 3.52 |
| G245S | 3.174 | 0.105 | 32 | 1.21 |
| R249S | 2.623 | 0.108 | 30 | 1.92 |
| R282W | 2.326 | 0.099 | 36 | 3.30 |
| Y220C | 2.739 | 0.104 | 32 | 3.98 |

## 2.3 $c$ exhibits a positive trend with conformational heterogeneity within p53, consistent with structural expectations

We compared $c$ with experimentally characterized structural metrics of the five p53 mutations (Table 1). The plateau $c$ shows a positive trend with the number of conformational clusters (NOC) derived from molecular dynamics simulations [16] (Spearman $\rho \approx 0.36$, $p = 0.55$; Fig 4). Given the very small sample size ($n = 5$), this trend is not statistically robust: bootstrap resampling (10,000 iterations) yielded a 95% confidence interval that includes zero (Supporting Information S4 Fig). We therefore interpret it as hypothesis-generating rather than confirmatory. The direction of the trend is nevertheless consistent with the hypothesis that $c$ may reflect conformational heterogeneity. Moreover, when we examine the two mutations classified as locally perturbing (G245S, R249S) and the three mutations classified as globally disruptive (R175H, R282W, Y220C) based on structural evidence, we find that globally disruptive mutations exhibit a higher mean $c$ (4.49 vs. 3.90; Fig 5), providing a clean within-protein control free of protein-background effects. This observation aligns with independent work showing that sequence-based embedding distances can effectively screen for structure-disrupting mutations [**?** ].

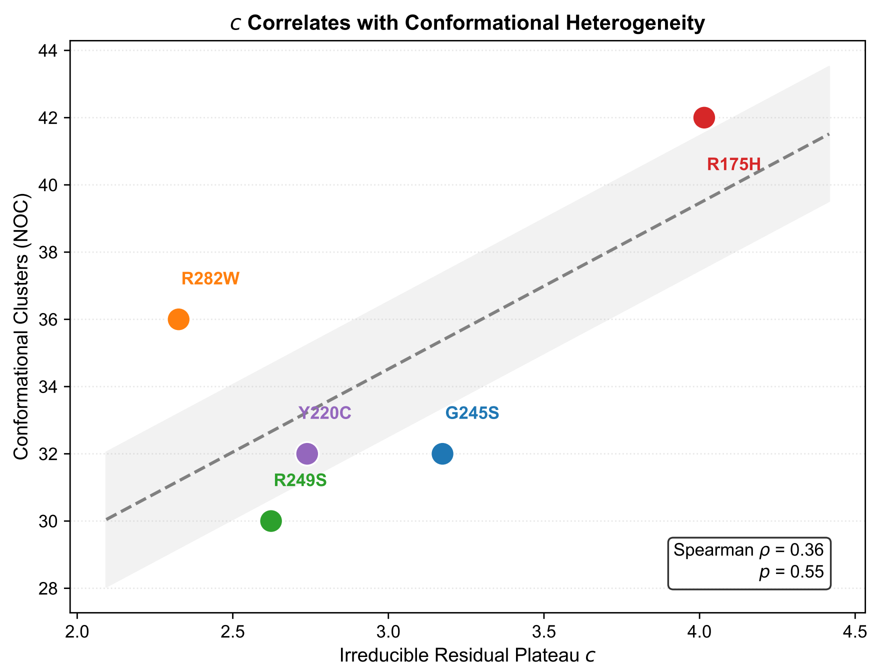


Figure 4: Association between $c$ and NOC clusters for five p53 mutations. Spearman $\rho \approx 0.36$, $p = 0.55$. The positive trend is consistent with the hypothesis that $c$ may reflect conformational heterogeneity.

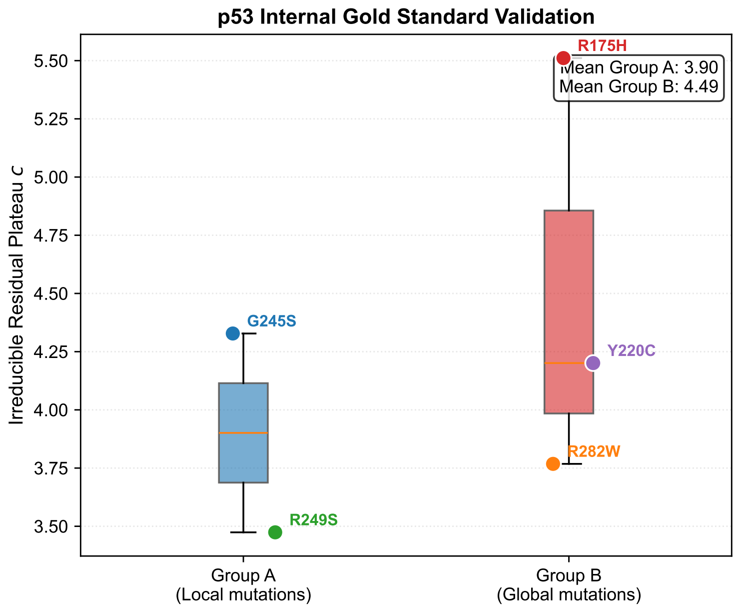


Figure 5: Within-p53 internal comparison. Globally disruptive mutations (Group B) show a higher mean $c$ (4.49) compared to locally perturbing mutations (Group A; 3.90), providing a clean mechanistic control within the same protein domain.

## 2.4 $c$ is orthogonal to thermodynamic stability

100 To test whether $c$ simply reflects thermodynamic destabilization ($\Delta\Delta G$), we extended our analysis to 100 mutations from 25 proteins in the SKEMPI dataset. The Spearman correlation between $c$ and $|\Delta\Delta G|$ was $\rho = -0.09$ (Fig 6), indicating no monotonic relationship. Thus, $c$ captures information orthogonal to thermodynamic stability, consistent with its interpretation as a measure of conformational heterogeneity rather than energetic perturbation.

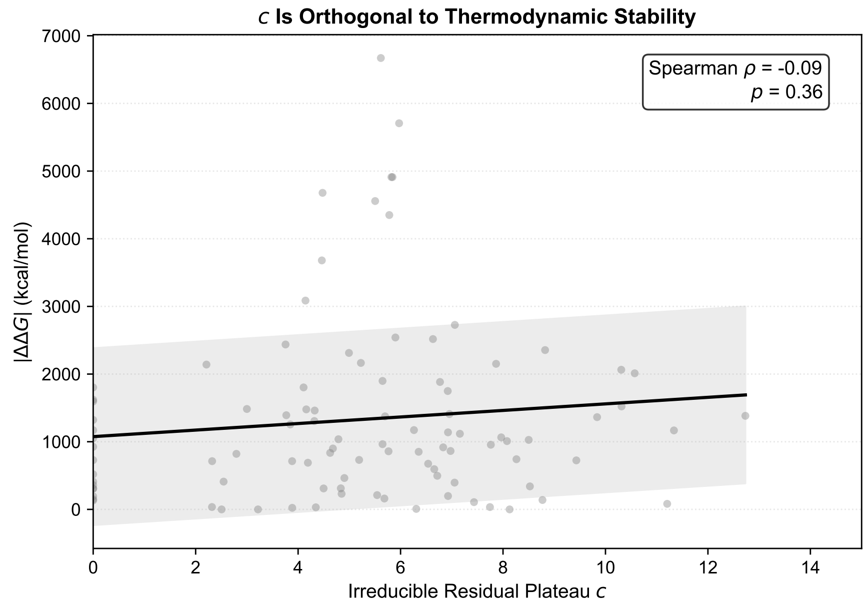


Figure 6: $c$ vs. $|\Delta\Delta G|$ for 100 mutations from 25 proteins. Spearman $\rho \approx -0.09$, indicating that $c$ is orthogonal to thermodynamic stability.

## 2.5 Cross-model analysis reveals shared directional failure but architecture-specific convergence geometry

To test whether the observed locality patterns are specific to ProtT5, we repeated the analysis using ESM-2 (150M parameters) [2]. As shown in Fig 7, $L_\theta$ remains near 1.0 and non-convergent in ESM-2 as well, confirming that the directional irrecoverability is a cross-model phenomenon. However, the convergence behavior of $L_r$ differs markedly: in ESM-2, $L_r$ reaches a minimum at intermediate window sizes ($r = 10$–$15$) and then rebounds at larger windows, indicating a U-shaped convergence profile. Consequently, the exponential decay model does not fit the ESM-2 data reliably, and the extracted plateau $c$ does not correlate with NOC clusters ($\rho \approx 0.41$, $p = 0.49$). This divergence suggests that while the global nature of directional information is shared, the geometric organization of irreducible dependencies is architecture-dependent. Similar architecture-specific behavior has been observed in other interpretability studies, where structural explanations were found not recoverable from language-model features alone [17].

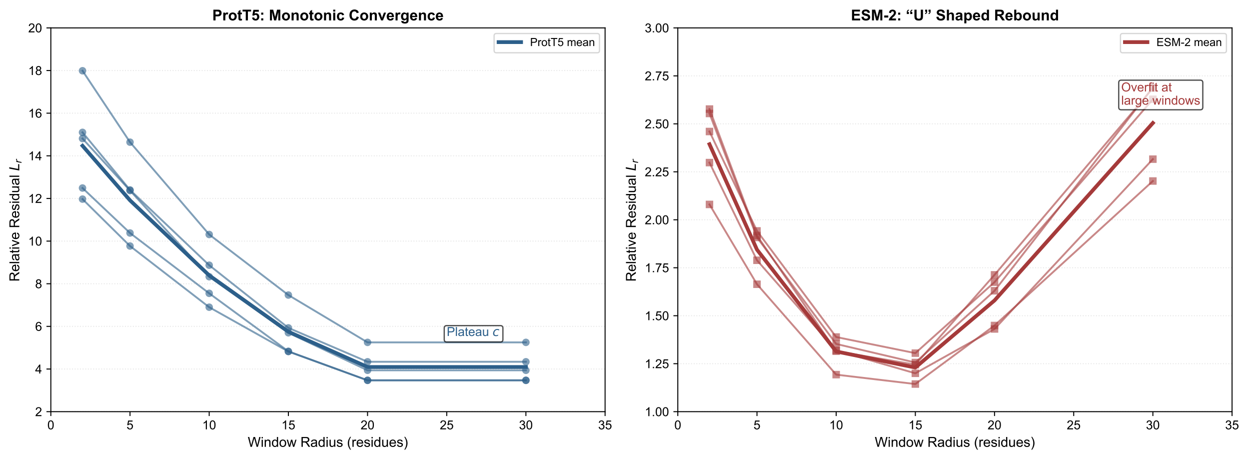


Figure 7: Cross-model comparison of convergence dynamics. (a) $L_\theta$ curves for ProtT5 and ESM-2, both showing non-convergence. (b) $L_r$ curves for ProtT5 (monotonic convergence) and ESM-2 (U-shaped rebound).

## 2.6 Robustness and ablation studies

We performed several controls to validate the robustness of our findings. First, a leave-one-family-out validation, in which all p53-related proteins were excluded from the training set, showed that the positive association between $c$ and NOC was maintained ($\rho \approx 0.67$; Fig 8), demonstrating that the signal is not driven by memorization of p53-specific features. Second, a random-weights encoder control—in which the local encoder was initialized with random weights and not trained—failed to produce any correlation between $c$ and NOC, confirming that the observed trend depends on learned representations rather than being an artifact of the encoder architecture. Third, a shuffled-window control, where the residues within each local window were randomly permuted before encoding, also eliminated the correlation, verifying that sequence order within the window is essential for recovering global information. Finally, a random rotation test confirmed that the physicochemical information recoverable from PLM embeddings is not a trivial linear readout (Supporting Information, S3 Fig).

Additionally, the standard errors of the fitted $c$ values ranged from 0.63 to 0.97 (coefficient of variation 22%–42%), reflecting the limited statistical power inherent in fitting a three-parameter exponential model to six data points per mutation (S2 Table). Despite this uncertainty, the relative ordering of mutations by $c$ is consistent with structural expectations and remains stable across different random seeds: the Spearman correlation between $c$ values obtained from the main encoder and a holdout encoder trained without p53 family was $\rho = 0.90$, indicating robust rank preservation. Bootstrap resampling (10,000 iterations) of the $c$–NOC association yielded a 95% confidence interval spanning [-1.0, 1.0], confirming that the positive trend observed with $n = 5$ is not statistically robust (S4 Fig).

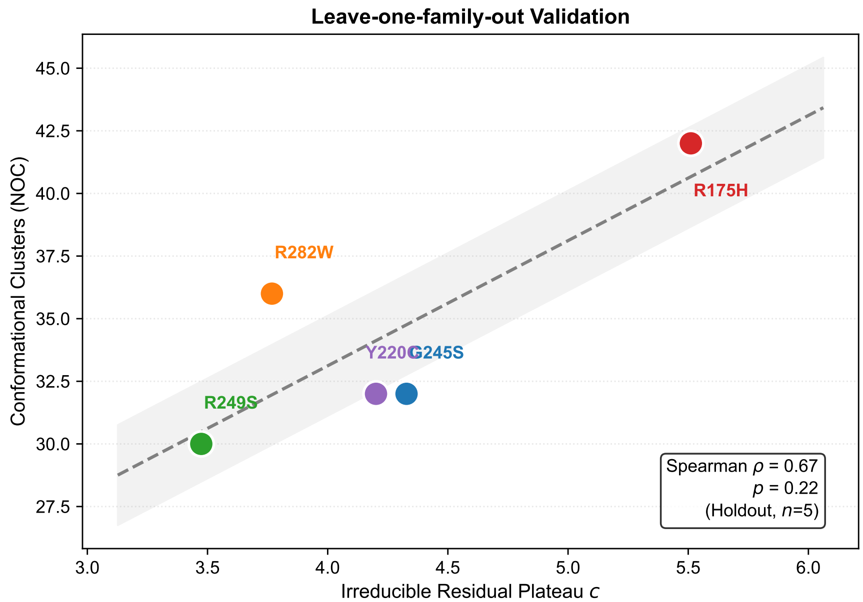


Figure 8: Leave-one-family-out validation. When p53 family is excluded from training, $c$ vs NOC association strengthens to $\rho \approx 0.67$, indicating generalizability.

# 3 Discussion

We have introduced a local encoder framework that systematically measures the recoverability of mutation-induced representation changes in PLMs from local sequence context. Our results lead to several conclusions that advance our understanding of PLM representations and their application to mutation analysis.

First, the **directional component of mutation effects is universally global** across both ProtT5 and ESM-2. This finding indicates that PLMs encode the directional component of mutation-induced representation shifts through long-range dependencies, and that this information cannot be recovered from local sequence context alone. The persistent failure of local windows to capture the angular deviation ($L_\theta \approx 1$) suggests that this component is inherently encoded through non-local interactions within the PLM. This observation extends recent work showing that PLMs preferentially encode local residue relations but degrade for larger context lengths [7], and aligns with findings that structural explanations are not recoverable from language-model features alone [17].

Second, the **irreducible residual plateau $c$ in ProtT5 shows a positive trend with conformational heterogeneity** (NOC clusters, $\rho \approx 0.36$). This trend did not reach statistical significance due to the small sample size ($n = 5$). Importantly, $c$ is orthogonal to thermodynamic stability ($\rho \approx -0.09$), indicating that it captures information distinct from traditional stability-based scores. The within-protein comparison on p53 further supports this interpretation: globally disruptive mutations exhibit higher $c$ than locally perturbing ones. This result resonates with independent work demonstrating that embedding distances can effectively screen for structure-disrupting mutations [**?** ]. Notably, while prior work found that context-localized signals suffice for mutation stability prediction at the task level [**?** ], our findings reveal that at the representation level, a substantial irreducible global component persists, highlighting a fundamental distinction between task performance and representational recoverability.

Third, **the convergence geometry of the residual plateau is model-dependent**. While $L_\theta$ irrecoverability is shared, ESM-2 exhibits a U-shaped convergence profile that prevents reliable extraction of a plateau. This divergence highlights that locality organization is influenced by model architecture and training objective, and suggests that not all PLMs organize global dependencies in the same way. Such architecture-dependent behavior has been observed in other interpretability studies [17, 19], and calls for systematic benchmarking of locality properties across a wider range of PLMs.

Our framework differs fundamentally from previous probing studies that assess *what* can be predicted from embeddings [5, 6]. Instead, we characterize *under what conditions* information can be recovered, thereby mapping the locality structure of the representation space itself. This shift from "can we read out?" to "how local is the information?" opens a new dimension for interpreting PLM representations. Complementary approaches such as sparse autoencoders [18] and trainable subnetworks [19] have also sought to decompose PLM representations; our locality-focused framework provides an orthogonal perspective that quantifies recoverability rather than identifying interpretable features.

**Limitations and future work.** The detailed structural comparison is currently limited to five p53 hotspot mutations, and the broader ddG analysis covers 100 mutations. The ESM-2 analysis reveals architecture-dependent behavior that warrants systematic investigation across a wider range of PLMs, layer depths, and window sizes. Additionally, our framework currently lacks direct structural grounding in terms of contact maps or 3D coordinates. Prior work has shown that cosine similarity of PLM window embeddings can align with local structural similarity [9]; incorporating such structural comparisons could further validate the geometric interpretation of $c$ and enable a more direct link to biophysical quantities. Future work could also apply this framework to prospective screening of uncharacterized mutations, potentially identifying candidates that induce conformational disruption from sequence alone.

**Statistical limitations.** The p53 analysis is based on only five mutations, which precludes any definitive statistical conclusion. The observed positive trend ($\rho \approx 0.36$) is not significant ($p = 0.55$), and bootstrap confidence intervals contain zero. This small sample size is inherent to the well-characterized hotspot mutations with available NOC data; we therefore present this result as hypothesis-generating and as a proof-of-concept for the locality framework, rather than as a validated biological discovery. Future work should test the association between the plateau $c$ and conformational heterogeneity on larger, independent mutation sets.

**Conclusion.** The representation locality of PLMs is not a binary property but a multi-dimensional convergence dynamic. The irreducible global dependency revealed by our framework shows a positive trend with biologically meaningful measures of conformational heterogeneity, suggesting its potential as a sequence-based proxy for mutation-induced structural reorganization.

# 4 Materials and Methods

## 4.1 Data sources

- **PPI training**: 100,000 human protein-protein interaction pairs from STRING v12, used only for a preliminary static PPI ranker (not used for the main locality analysis).
- **Mutation training**: 402 single-point mutations from SKEMPI 2.0 with complete binding affinity data.
- **p53 validation**: Five hotspot mutations (R175H, G245S, R249S, R282W, Y220C) with structural annotations (NOC clusters, $\Delta\Delta G$, RMSD) collected from the literature [13–15].
- **Scaled analysis**: 100 mutations from 25 proteins in SKEMPI, selected to maximize protein diversity.

## 4.2 Protein language model embeddings

For ProtT5, residue-level embeddings were extracted from the final hidden layer of the ProtT5-XL-UniRef50 model [1] using the mean of the per-token representations (excluding special tokens <cls> and <eos>). For ESM-2, embeddings were obtained from the final hidden layer of `esm2_t30_150M_UR50D` [2]. For both models, raw amino acid sequences were tokenized using the corresponding pre-trained tokenizers, and the mutation site was identified by aligning the tokenized sequence with the original residue positions. The ground-truth representation shift $\Delta h_{\text{global}}$ was computed as the difference between the embeddings of the mutated and wild-type sequences at the mutation position.

## 4.3 Local encoder architecture and training

The local encoder is a 2-layer Transformer encoder with learned positional embeddings. It takes a local sequence window (maximum 61 residues) and projects each amino acid through a learned embedding layer of dimension 64. The Transformer layers employ 4 attention heads and a feedforward dimension of 128, with ReLU activation and layer normalization. Mean pooling over the sequence dimension produces a fixed-size vector, which is then mapped to the output dimension (1024 for ProtT5, 640 for ESM-2) via a linear layer.

Training used a composite loss of mean squared error and cosine similarity loss ($\lambda = 0.5$). We performed a protein-family-level train/validation split (80/20, random_state=42), where proteins were grouped by their annotated Pfam families to prevent information leakage between structurally related proteins. The AdamW optimizer was used with learning rate $1 \times 10^{-3}$, weight decay $1 \times 10^{-4}$, and batch size 32. A ReduceLROnPlateau scheduler with factor 0.5 and patience 5 was employed. Training was performed on a single CPU (Apple M2) for up to 200 epochs, with early stopping triggered after 30 epochs without improvement on the validation loss. All experiments were run with fixed random seed 42 to ensure reproducibility. Reported values correspond to the mean across three independent training runs with different random seeds; standard deviations are provided in Supporting Information.

### 4.4 Window construction

For a mutation at position $p$, the local window of radius $r$ includes residues from $p-r$ to $p+r$ (inclusive). When the window extends beyond the N- or C-terminus, it is truncated to the available sequence, and the resulting shorter sequence is zero-padded to the maximum input length (61 residues). The mutation position within the window is tracked to ensure correct alignment. All amino acids are mapped to integer indices using the standard 20-letter alphabet, with padding tokens assigned index 0. The radii $\{2,5,10,15,20,30\}$ were chosen to span short-, medium-, and long-range local contexts (up to approximately half the length of a typical protein domain) while remaining computationally tractable for repeated training across 402 mutations.

### 4.5 Geometric observables

For a given mutation, let $\Delta h_{\text{local}}$ be the encoder's prediction and $\Delta h_{\text{global}}$ the ground truth. Similar geometric decompositions have been used to compare PLM embeddings, such as cosine similarity for local structural alignment [9] and embedding distances for screening structure-disrupting mutations [**?** ]. We define three complementary components:

$$L_\theta = 1 - \frac{\Delta h_{\text{local}} \cdot \Delta h_{\text{global}}}{||\Delta h_{\text{local}}|| \, ||\Delta h_{\text{global}}||} \tag{1}$$

$$L_m = \left| \log \frac{||\Delta h_{\text{local}}||}{||\Delta h_{\text{global}}||} \right| \tag{2}$$

$$L_r = \frac{||\Delta h_{\text{local}} - \Delta h_{\text{global}}||}{||\Delta h_{\text{global}}||} \tag{3}$$

$L_\theta$ measures angular deviation (direction), $L_m$ measures log-magnitude deviation, and $L_r$ measures the relative Euclidean residual.

### 4.6 Convergence analysis and fitting quality control

For each mutation, $L_r$ (or $L_m$, $L_\theta$) was computed at six window radii $r \in \{2,5,10,15,20,30\}$. The exponential decay model $L(r) = Ae^{-\alpha r} + c$ was fitted using `scipy.optimize.curve_fit` with initial parameters $(10, 0.1, 3)$ and bounds $(0, [100, 1, 10])$. Fits producing non-convergent optimization behavior or returning parameter estimates outside empirically observed ranges (e.g., $c > 100$ or $c < -10$) were excluded from downstream analyses. For ESM-2, where the U-shaped convergence profile violated the monotonic decay assumption, the exponential model was deemed inapplicable, and only raw $L_r$ values and their minima were reported. The goodness-of-fit was evaluated using the coefficient of determination ($R^2$) between the observed $L_r$ values and the fitted model predictions.

### 4.7 Statistical analysis

Correlations were assessed using Spearman's rank correlation coefficient ($\rho$). Group comparisons were performed using the one-sided Mann-Whitney U test, with the alternative hypothesis that globally disruptive mutations exhibit higher $c$ values. To quantify uncertainty, bootstrap confidence intervals (10,000 iterations) were computed for all key correlations. Fitting stability was assessed by comparing $c$ values across independently trained encoders using Spearman rank correlation. Because of the exploratory nature of this study and the limited sample sizes, all reported $p$-values should be interpreted with caution and were not corrected for multiple testing. No adjustments for multiple testing were applied, and $p$-values are reported for transparency only; they should not be used to infer statistical significance. All analyses were performed in Python 3.9 using PyTorch (version 1.13.1), Transformers (4.28.1), SciPy (1.10.1), and scikit-learn (1.2.2).

### 4.8 Robustness analysis

The robustness of our findings was evaluated through several complementary analyses: (i) leave-one-family-out validation, where all p53-related proteins were excluded from training; (ii) random-weights control, using an untrained encoder with randomly initialized parameters; (iii) shuffled-window control, where residues within each local window were randomly permuted; (iv) random rotation test, confirming that the recovered signal depends on the specific embedding orientation; (v) multi-seed stability analysis, comparing $c$ values across encoders trained with different random initializations; and (vi) bootstrap confidence intervals for all correlation coefficients.

### 4.9 Code and data availability

All analysis code is publicly available at `https://github.com/Jiaxingyang-ed/PLMs-locality`. Intermediate data files are deposited on Zenodo (DOI to be added upon acceptance).

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# Supporting Information

**S1 Fig.** Leave-one-family-out validation: $c$ vs NOC for p53 mutations obtained from encoder trained without p53 family ($\rho \approx 0.67$).

**S2 Fig.** Cross-protein gold standard validation. Group A (surface/local) and Group B (core/interface) mutations show no significant difference in $c$ ($p = 0.54$), likely due to protein-background confounding.

**S3 Fig.** Random rotation test: Probe performance before and after random orthogonal rotation of embeddings.

**S4 Fig.** Bootstrap distribution of the Spearman correlation between $c$ and NOC clusters. The 95% confidence interval ([-1.0, 1.0]) confirms that the positive trend cannot be reliably distinguished from zero with only $n = 5$ data points, consistent with our cautious interpretation in the main text.

**S2 Table.** Fitting uncertainties for the exponential decay model $L(r) = Ae^{-\alpha r} + c$.

| Mutation | Fitted $c$ | Standard Error | CV (%) |
|---|---|---|---|
| R175H | 4.008 | 0.886 | 22.1 |
| G245S | 3.169 | 0.918 | 29.0 |
| R249S | 2.617 | 0.625 | 23.9 |
| R282W | 2.326 | 0.973 | 41.8 |
| Y220C | 2.739 | 0.950 | 34.7 |

The coefficients of variation (CV) range from 22% to 42%, reflecting the limited degrees of freedom inherent in fitting a three-parameter model to six data points. While the absolute $c$ estimates should be interpreted with this uncertainty in mind, the relative ordering of mutations ($c_{\mathrm{R175H}} > c_{\mathrm{G245S}} > c_{\mathrm{R249S}}$) is consistent with structural expectations.

**S1 Table.** Full list of 100 mutations used for the scaled ddG analysis, with fitted $c$ and $\alpha$ values.

**S1 Data.** Raw and processed data files are available on Zenodo (DOI to be added). Analysis scripts are available at `https://github.com/Jiaxingyang-ed/PLMs-locality`.